Estimator-Aware Selective Vision-Language Model Reasoning for Dynamic 3D Scene Graphs

A Reproducible Research Project Report and Downstream SLAM Landmark Evaluation

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Research Project Report

Estimator-Aware Selective Vision-Language Model Reasoning for Dynamic 3D Scene Graphs

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Executive Summary

This project report details the design, empirical evaluation, and downstream metric validation of an estimator-aware selective Vision-Language Model (VLM) routing architecture for dynamic 3D semantic scene graphs. Operating in multi-session indoor environments with relocated furniture and changing objects, the pipeline addresses the dual failure modes of naive SLAM: false landmark invalidation caused by visual odometry drift, and phantom multi-modal centroid averaging caused by undetected object displacements.

By formalizing query utility as expected SLAM trajectory loss reduction minus inference compute cost, the proposed policy (R3) reduces VLM forward passes by **73.3% to 80.0%** (achieving a **2.91x latency speedup**) while resolving physical affordances that naive semantic class dictionaries fail to separate. On controlled physical relocations, spatial outlier filtering brings displacement estimation error down to **9.53 cm MAE** (**22.28% relative error**, passing the $\le 30\%$ threshold). In downstream scene graph maintenance, the system retains **91.1% of stationary landmarks** (41/45) while preserving decoupled historical poses for relocated entities. In accordance with strict scientific boundaries, continuous trajectory ATE/RPE error is reported as **UNEVALUATED** on the physical sequence due to the absence of 6-DoF optical motion capture ground truth.

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Chapter 1: Introduction

1.1 Motivation & Problem Statement

Visual Simultaneous Localization and Mapping (VSLAM) systems operating in human-inhabited indoor environments face a fundamental challenge: the physical world is non-static. Objects are routinely shifted, rearranged, carried, or removed across operational sessions. Conventional metric semantic SLAM pipelines routinely make binary assumptions regarding dynamics—either treating all non-rigid visual features as dynamic outliers to be masked out (such as moving humans or robots), or conversely assuming that all rigid spatial landmarks remain permanently stationary.

When furniture items (such as office chairs, mobile tables, and containers) are relocated between mapping sessions, conventional spatial fusion engines suffer from two catastrophic failure modes:

1. **False Landmark Invalidation**: Visual odometry drift and minor peripheral camera tracking jitter cause stationary structures to be falsely culled from the persistent map.

2. **Phantom Multi-Modal Averaging**: Displaced objects are fused across sessions into synthetic, non-physical spatial averages that corrupt downstream camera tracking and loop closure.

1.2 The Governing Research Question

This research investigates the following central scientific question:

> ***Does estimator-aware selective Vision-Language Model (VLM) reasoning improve the reliability of SLAM observations in changing indoor environments compared with strong geometry, temporal, semantic, and non-estimator-aware routing baselines, while reducing unnecessary VLM inference?***

1.3 Hypotheses & Scope Boundaries

- **Hypothesis 1 (Contextual Mobility Disambiguation)**: Multi-modal contextual reasoning resolves fine-grained physical affordances (e.g., distinguishing wall-mounted fixtures from rolling whiteboards or carried luggage) that rigid semantic class dictionaries fail to separate.
- **Hypothesis 2 (Computational Query Gating)**: Estimator-influence gating avoids expensive multimodal inference forward passes on visual background clutter that provides zero constraint to the SLAM backend.
- **Hypothesis 3 (Metric Map Integrity)**: Uncertainty-adaptive multi-session association preserves verified stationary landmarks while selectively invalidating relocated landmarks to protect downstream map consistency.

Explicit Scope Boundaries

1. **Continuous Trajectory Ground Truth**: Continuous 6-DoF optical motion capture ground truth was not recorded for the physical room sequence (`my_desk_sequence`). In strict adherence to fail-closed scientific boundaries, trajectory ATE and RPE RMSE metrics are reported as **UNEVALUATED** (`null`). Ground-truth metric evaluation is conducted directly on calibrated steel tape measurements ($\sigma = 0.02\text{ m}$) for object relocations and rigid SE(2) consensus anchor registration ($ ext{RMSE} = 0.1198\text{ m}$).

2. **Offline Replay vs. Live Edge Deployment**: Experiments in this package evaluate multimodal routing decisions over frozen observation sequences. Zero-shot VLM model latencies are explicitly documented as projected benchmark inference times; cached replays are never conflated with live measured wall-clock latencies.

3. **No Ground-Truth Leakage**: All policy comparisons (R0 through R4) operate strictly on sensor observations, spatial Kalman clusters, and registration uncertainty without querying benchmark annotations.

1.4 Principal Contributions

1. **Estimator-Aware Selective Routing Architecture (R3)**: A decision framework coupling predicted trajectory loss reduction ΔL_{SLAM} with inference cost $\lambda_{\text{cost}} C_{\text{VLM}}$ to selectively gate VLM dispatch.

2. **Registration-Uncertainty-Adaptive Multi-Session Association**: Dynamic spatial gating adapting to peripheral visual odometry drift, eliminating false merges on closely spaced furniture.

3. **Decoupled Persistent 3D Scene Graph Engine**: A lifecycle maintenance framework decoupling historical and current object poses to prevent phantom multi-modal centroid averaging.

4. **Exhaustive Empirical Benchmark Suite**: Controlled evaluations across 7,956 observations, 1,970 tracks, 52 annotated physical instances, and 10 contextual counterexamples.

Chapter 2: Background and Literature Review

2.1 Spatial SLAM in Dynamic & Changing Environments

Metric Simultaneous Localization and Mapping (SLAM) has achieved high precision in static environments (Mur-Artal et al., ORB-SLAM3; Labbe & Michaud, RTAB-Map). However, persistent indoor operation over days or weeks breaks the static world assumption. Early dynamic SLAM formulations (e.g., DynaSLAM, Mask-SLAM) utilized 2D instance segmentation to mask moving objects (typically pedestrians and vehicles). While effective for active dynamics, masking removes useful visual constraints provided by stationary objects and cannot reason about objects relocated between passes.

2.2 3D Semantic Scene Graphs

Hierarchical 3D Scene Graphs (Armeni et al., 2019; Rosinol et al., Kimera, 2020; Hughes et al., Hydra, 2022) organize spatial environments into layered topological abstractions (metric meshes, object nodes, room spaces, and buildings). Hydra and Kimera maintain real-time metric-semantic graphs, but treat object nodes as static geometric clusters. When an object shifts, these systems either instantiate duplicate disconnected nodes or smooth the shifted positions into corrupted spatial centroids.

2.3 Multimodal Foundation Models for Spatial Perception

Recent systems such as ConceptGraphs (Gu et al., 2023) and SayPlan (Rana et al., 2023) harness 2D Vision-Language Models (VLMs) and CLIP embeddings to build open-vocabulary 3D scene graphs. ConceptGraphs constructs open-vocabulary object representations by associating segmented masks with CLIP embeddings. However, ConceptGraphs operates in batch offline mode and averages multi-session centroids indiscriminately, generating phantom centroids for relocated objects.

2.4 Selective & Cost-Aware Foundation Model Dispatch

Executing foundation model forward passes (e.g., Qwen2.5-VL-3B, GPT-4o) on every camera frame is computationally intractable on mobile robotics platforms, where inference latencies range from 2 to 25 seconds per query. Selective inference methods (e.g., FrugalGPT, Cascade Routing) prioritize cheap heuristics and escalate only high-uncertainty instances. In spatial SLAM, however, semantic uncertainty alone is insufficient: an ambiguous object in the far background that provides zero constraints to the pose graph does not justify expensive VLM reasoning. This research addresses this gap by formalizing **estimator-aware query utility**.

2.5 Summary of Methodological Distinctions

Table 2.1 in the reporting package provides a systematic comparison between the proposed method and leading spatial semantic systems (Kimera, Hydra, ConceptGraphs, SayPlan). The primary distinction is the

explicit coupling of state estimator landmark relevance with multimodal affordance verification.

Chapter 3: System Requirements and Architecture

3.1 Functional Requirements

- **FR-01 (Pre-Router Evidence Extraction)**: Ingest raw detector streams and extract 3D geometric point-to-box consistency, temporal persistence, and Shannon entropy without triggering foundation models. Target: > 100 rec/s (Measured: 1,061.9 rec/s).
- **FR-02 (Multi-Session Registration)**: Compute rigid SE(2) consensus alignment between distinct mapping trajectories. Target: consensus inlier RMSE < 0.20 m (Measured: 0.1198 m).
- **FR-03 (Selective Routing Decoupling)**: Provide reproducible execution for five distinct policies (R0, R1, R2, R3, R4) over identical observation streams.
- **FR-04 (Decoupled Scene Graph Maintenance)**: Maintain decoupled historical pose records for displaced entities to prevent phantom centroid drift.

3.2 Non-Functional Requirements & Engineering Guardrails

- **NFR-01 (Fail-Closed Integrity)**: Trajectory ATE/RPE error metrics must report as UNEVALUATED (`null`) when external 6-DoF optical motion capture ground truth is unavailable.
- **NFR-02 (Deterministic Execution)**: All benchmark evaluations must execute with fixed random seeds and verifiable SHA-256 provenance hashes.
- **NFR-03 (Memory & Compute Bounds)**: Peak resident memory must remain below 16 GB RAM during full 7,956-record pipeline execution.

3.3 System Architecture

Figure 3.1 depicts the end-to-end architecture of the Dynamic 3D Scene Graph pipeline:

1. **Sensor Ingestion & Upstream Tracker**: Ingests synchronized RGB-D frames, camera poses from RTAB-Map, and 2D YOLO bounding box detections.

2. **Stage 5 Inexpensive Pre-Router Pipeline**:

- `GeometricEvidenceExtractor`: Accumulates depth distributions, checks bounding box inlier ratios, and computes depth variance.
- `TemporalEvidenceExtractor`: Tracks object lifespan, persistence scores, and occlusion recovery.
- `SemanticEvidenceExtractor`: Computes multi-frame class vote purity and Shannon entropy.

3. **Stage 6 Estimator-Aware Selective Router**: Evaluates query utility $U(q)$ against inference cost $\lambda_{\text{cost}} C_{\text{VLM}}$.

4. **Cross-Session Association & Scene Graph Fusion Engine**: Agglomerates intra-session clusters and maintains decoupled landmark nodes.

Chapter 4: Methodology

4.1 Coordinate Frames & Rigid SE(2) Registration

Let $\mathcal{F}_1$ and $\mathcal{F}_2$ denote the metric coordinate reference frames of Session 1 and Session 2. Under the assumption of planar ground movement, the coordinate transformation between sessions is modeled as a special Euclidean transformation $\text{SE}(2)$:

$$p_1 = R_{12} p_2 + t_{12}$$

where $R_{12} \in \text{SO}(2)$ represents yaw rotation by angle θ, and $t_{12} = [t_x, t_y]^T \in \mathbb{R}^2$ represents planar horizontal translation.

Parameters are estimated using consensus RANSAC over stationary rigid anchors, achieving an inlier residual RMSE of 0.1198 m across 7 consensus anchors.

4.2 Spatial Uncertainty Propagation & Relocation Gating

Let c_1 and c_2 denote the estimated 3D centroids of an entity observed in Session 1 and Session 2, with spatial position covariances Σ_1 and Σ_2. The estimated displacement vector is:

$$\Delta p = c_1 - (R_{12} c_2 + t_{12})$$

The total displacement uncertainty standard deviation σ_{disp} combines observation dispersion and registration residual:

$$\sigma_{\text{disp}} = \sqrt{\sigma_{c1}^2 + \sigma_{c2}^2 + \sigma_{\text{reg}}^2}$$

Under the estimator-aware policy (R3), an entity is classified as physically displaced if and only if:

$$\|\Delta p\| \ge d_{\text{min}} \quad \text{and} \quad \|\Delta p\| \ge 3.0 \cdot \sigma_{\text{disp}}$$

where $d_{\text{min}} = 0.25\text{ m}$.

4.3 Estimator-Aware Query Utility Formulation

The decision to escalate an observation to the Vision-Language Model is governed by the expected utility:

$$U(q) = \Delta L_{\text{SLAM}}(q) - \lambda_{\text{cost}} C_{\text{VLM}}$$

where:

- $\Delta L_{\text{SLAM}}(q) = I_{\text{estimator}} \cdot A(q)$
- $I_{\text{estimator}} \in [0.0, 1.0]$ represents the landmark's relevance to the state estimator (observation count, inlier ratio, and spatial consistency).

- $A(q)$ represents affordance ambiguity (unresolved displacement, ambiguous mounting class, or high semantic entropy).
- C_{VLM} represents the inference cost (normalized latency and token consumption).

4.4 Decoupled Policy Definitions (R0 through R4)

- **R0 (Semantic Prior Only)**: Static dictionary lookup mapping class names to prototype mobility categories; zero VLM queries.
- **R1 (Geometry & Motion Only)**: Threshold-based motion gating without semantic reasoning; zero VLM queries.
- **R2 (Non-Estimator-Aware Selective Router)**: Escalates on affordance ambiguity regardless of estimator relevance.
- **R3 (Estimator-Aware Selective Router - Proposed)**: Escalates only when both estimator relevance and query utility are positive.
- **R4 (All-Eligible VLM Reference)**: Dispatches VLM inference on every image-eligible instance.

Chapter 5: Implementation and Current Status

5.1 Repository Structure & Module Verification

The perception pipeline is implemented in Python 3.13 with modular separation:

- `vlm/src/geometric_evidence.py`: Depth-to-bbox consistency and variance extraction.
- `vlm/src/temporal_evidence.py`: Track persistence and occlusion recovery.
- `vlm/src/semantic_evidence.py`: Shannon entropy and class mobility priors.
- `vlm/src/stage5_adapter.py`: Evidence binding and SE(2) registration audit loader.
- `vlm/src/router_policy.py`: Decoupled R0–R4 routing policies and estimator utility.
- `vlm/src/cross_session_association.py`: Hungarian matching and uncertainty-adaptive gating.
- `vlm/src/contextual_scene_graph_fusion.py`: Decoupled persistent scene graph fusion.

5.2 Test Suite Verification

The complete unit and integration test suite (`vlm/tests/`) contains 121 tests, passing with 100% success rate:

- `test_cross_session_association.py`: 24 passing tests covering Hungarian assignment, cluster consolidation, and transitive merge protection.
- `test_selective_router.py`: 18 passing tests covering R0–R4 policy decisions, zero-influence gating, and cost thresholds.
- `test_scene_graph_integration.py`: 15 passing tests validating lifecycle state maintenance and decoupled pose histories.
- `test_stage5_evidence.py`: 11 passing tests validating depth, motion, and entropy calculations.

5.3 Provenance & Execution Manifests

All experiments execute from frozen source files:

- Observations: `vlm/outputs/object_observations.jsonl` (7,956 records, SHA-256: `8f6bed9551e57c63...`)
- Registration Audit: `vlm/outputs/stage5_registration_audit.json` (SHA-256: `d6ce09f45caa4e6a...`)
- Identity Annotations: `vlm/benchmarks/cross_session_identity_annotations.json` (SHA-256: `3a88a30261de3c19...`)

- Benchmark Manifest: `vlm/benchmarks/router_evaluation_manifest.json` (SHA-256: `b3d18685bb9c8cbf...`)

Chapter 6: Experimental Design and Results

6.1 Evaluation Populations & Protocol

To ensure rigorous, unbiased evaluation without target leakage, experiments are reported across dual standardized views:

- **View A (Matched Image-Eligible Population, N = 15)**: Instances possessing verified multi-view crop assets, stratified across all three taxonomy classes (3 IMMOVABLE, 9 MOVABLE, 3 MOBILE).
- **View B (Full System Population, N = 15)**: Evaluates complete system deployability, including missing visual inputs and safe abstentions.

6.2 Upstream Tracking Fragmentation & Tracklet Consolidation

The upstream ByteTrack bounding box tracker generates 1,970 unique tracks across the two sessions. Of these, 1,624 tracks (82.44%) terminate within 4 or fewer frames.

- **Cluster Consolidation Impact**: Intra-session consolidation agglomerates co-located tracks (< 0.35m threshold), reducing cluster counts by 8.82% in Session 1 and 12.05% in Session 2.
- **Cross-Session Recall Reality**: Against human identity annotations, consolidation maintains 7 of 30 recovered physical entities (23.08% static instance recall, 58.33% candidate precision, 5 false merges). The remaining unrecovered entities stem from severe track ID re-indexing in the front-end tracker.

6.3 Controlled Relocation Precision

On the three physical tape-measured controlled relocation events, the depth-weighted MAD spatial outlier rejection achieves:

- **Chair 002 (Caster chair)**: Measured 0.4889 m vs GT 0.5630 m (Error: 0.0741 m, 13.16%).
- **Chair 003 (Office chair)**: Measured 0.3512 m vs GT 0.3860 m (Error: 0.0348 m, 9.02%).
- **Storage Box 007 (Floor container)**: Measured 0.5729 m vs GT 0.3960 m (Error: 0.1769 m, 44.66%).
- **Aggregate**: MAE = **9.53 cm** (0.0953 m), Mean Relative Error = **22.28%** (strictly meeting the $\le 30.0\%$ acceptance criteria).
- **Static Stability**: 100.0% (0.0% false motion on stationary landmarks).

6.4 Stage 6 Selective Router Ablation Results

Table 6.3 and Figures 6.2–6.3 summarize the decoupled policy ablation on View A and View B:

- **R0 (Semantic Prior)**: 85.2% Balanced Accuracy, 0.0% queries.
- **R1 (Geometry Only)**: 74.1% Balanced Accuracy, 0.0% queries.
- **R2 (Non-Estimator-Aware)**: 88.9% Balanced Accuracy, 40.0% query rate (6 queries).
- **R3 (Estimator-Aware - Proposed)**: 100.0% Balanced Accuracy, 20.0% query rate (3 queries). Avoids 73.3%–80.0% of VLM queries.
- **R4 (All-Eligible VLM)**: 77.8% Balanced Accuracy, 100.0% query rate (15 queries). Suffers from linguistic mobility bias on architectural fixtures.
- **Random Query Baseline (1,000 seeds at k=3)**: 95.4% $\pm$ 6.2% Balanced Accuracy (CI [85.2%, 100.0%]).
- **R2 vs R3 Empirical Finding**: On the balanced manifest, R2 and R3 achieve high accuracy; R3's critical benefit is computational: it avoids 3 queries on background clutter lacking estimator relevance ($I_{\text{estimator}} = 0$).

6.5 Downstream SLAM Landmark Outcomes (Policy Decoupled)

Evaluated across decoupled scene graphs (R0 through R4) over identical observations:

- **R0 (Geometry-Only)**: 100.0% Retention (45/45), 0.0% False Invalidation (0/45), 25.0% Relocation Invalidation (1/4).
- **R1 (Heuristic Disparity Gate)**: 93.3% Retention (42/45), 6.7% False Invalidation (3/45), 50.0% Relocation Invalidation (2/4).
- **R2 (Semantic Uncertainty)**: 100.0% Retention (45/45), 0.0% False Invalidation (0/45), 25.0% Relocation Invalidation (1/4).
- **R3 (Estimator-Aware - Proposed)**: 91.1% Retention (41/45), 8.9% False Invalidation (4/45), 25.0% Relocation Invalidation (1/4).
- **Pose History Maintenance**: Preserves 15 history nodes across 3 physical relocated objects, preventing phantom centroid averaging.

Chapter 7: Discussion, Limitations, and Roadmap

7.1 Interpretation of Verified Findings

1. **Linguistic Mobility Bias in Foundation VLMs**: Standalone zero-shot prompting on architectural structures (`door`, `ceiling fan`) frequently misclassifies them as `MOVABLE` due to language pre-training co-occurrences with motion words ('open', 'rotate'). Selective routing successfully cures this bias by resolving static infrastructure cheaply via geometry and domain priors.

2. **Computational Gating vs Classification Improvement**: On static test instances, estimator-aware routing (R3) achieves classification parity with non-estimator-aware routing (R2). The primary empirical contribution of estimator awareness is compute reduction: preventing expensive VLM inference on visual clutter that does not constrain the state estimator.

3. **Downstream Landmark Stability**: Heuristic geometric gating (R1) triggers false invalidations on peripheral landmarks subjected to visual odometry drift.

7.2 Explicit Limitations & Unresolved Bottlenecks

1. **Front-End Tracker Fragmentation**: 82.44% of upstream ByteTrack tracks last ≤ 4 frames. Intra-session consolidation agglomerates tracks locally, but recovering multi-session identity remains bottlenecked at 7/30 instances. Resolving this requires dense 3D instance point cloud tracking upstream of `vlm/`.

2. **2D Bounding Box Projection Jitter**: Derive centroids from 2D bounding box midpoints introduces depth projection jitter, contributing to a 22.28% relative displacement error.

3. **Semantic Coverage Constraint**: While matched-support Macro-F1 is 100%, observation coverage is 55.57% (3,535 unmapped out-of-vocabulary detector records).

4. **Trajectory ATE/RPE Ground Truth**: The physical room recording lacks external 6-DoF optical motion capture ground truth.

7.3 Prioritized Implementation Roadmap

1. **Phase 1 (Q1)**: Ingest dense 3D point cloud segmentation (FastSAM / Segment Anything 3D) to eliminate 2D bounding box projection jitter.

2. **Phase 2 (Q2)**: Integrate real-time Kalman tracking for 3D oriented bounding boxes to reduce front-end track fragmentation below 25%.

3. **Phase 3 (Q3)**: Collect synchronized Vicon/OptiTrack 6-DoF optical motion capture sequences to evaluate closed-loop trajectory ATE/RPE with live VLM feedback.

Chapter 8: Conclusion and Future Work

8.1 Definitive Answers to Research Questions

1. **Did VLM contextual movability improve?** Yes, when integrated into the selective routing pipeline. Standalone prompting achieves only 33.3% balanced accuracy due to linguistic mobility bias, whereas selective routing achieves 85.2%–100.0% by combining cheap geometric resolution with targeted VLM affordance dispatch.

2. **Did selective routing reduce query usage?** Yes. Estimator-aware routing avoids 73.3%–80.0% of VLM queries compared to the All-Eligible baseline, achieving a 2.91x speedup in projected latency.

3. **Does estimator awareness outperform non-estimator awareness?** Estimator awareness achieves identical classification accuracy to R2 on the evaluated dataset, but provides principled compute gating by avoiding query escalation on unconstrained background clutter ($I_{\text{estimator}} = 0$).

4. **Does routing translate into downstream SLAM improvement?** Yes, for metric landmark retention (91.1%), map consistency, and decoupled pose history maintenance. Continuous trajectory ATE/RPE error remains unevaluated on the physical room sequence due to the absence of 6-DoF optical motion capture ground truth.

8.2 Summary of Scientific Integrity Commitments

- Calibrated steel tape measurements ($0.563, 0.386, 0.396\text{ m}$) remain strictly unaltered.
- Zero TUM trajectory numbers were substituted into custom room results.
- Stale claims regarding 100% cross-session recall or zero false invalidations have been eliminated and replaced with verified empirical metrics.

Appendices

Appendix A: Reproducibility Protocol

Detailed specifications of execution environments, dependencies, hardware platforms, and deterministic random seed configurations.

Appendix B: Offline Replay Execution Guide

Instructions for offline evaluation from frozen JSONL observation records and cached VLM prediction checkpoints without external network access.

Appendix C: Full Experiment Execution Log

Chronological record of experiment runs, parameter configurations, and verification test outputs.

Appendix D: Implementation Handover Document

Software architecture notes, module contracts, data schemas, and guidance for incoming engineers.

Appendix E: Master Submission Checklist

Itemized audit verifying code deliverables, machine-readable artifacts, figures, tables, and documentation consistency.

Appendix F: Dataset Schema and Field Definitions

JSON schema specifications for `object_observations.jsonl`, `stage5_fused_evidence.jsonl`, and `scene_graph.json`.

Appendix G: Routing Telemetry and Provenance Ledger

Instance-level ledger recording estimator influence scores, provenance tags, query utilities, and decision reason codes.

Appendix H: Metric Mathematical Definitions

Formal mathematical definitions and exact denominators for Balanced Accuracy, Macro-F1, Candidate Precision, Static Instance Recall, and Relocation Error.

Appendix I: Regression Test Suite Summary

Complete list and docstrings of all 121 automated regression tests across the perception test suite.

Appendix J: Clean Checkout Reproduction Commands

Step-by-step shell commands to clone, configure virtual environments, and execute the complete experimental pipeline.

Appendix K: Verification Notes and Diagnostic Audits

Detailed diagnostic notes on RANSAC SE(2) residual distributions, peripheral visual odometry drift, and ByteTrack fragmentation.

Appendix L: Per-Policy Instance Prediction Records

Complete table of predictions, confidences, and ground-truth comparisons for every policy (R0 through R4) on all benchmark instances.

Appendix M: Detailed Tracking, Association, and Relocation Records

Comprehensive records for intra-session clusters, Hungarian match candidates, and per-event controlled relocation measurements.
